



LIABILITY-ADJUSTED BOND PORTFOLIO CONSTRUCTION IN THE SRI LANKAN GOVERNMENT SECURITIES MARKET

A QUANTITATIVE APPROACH



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1) Introduction

This report presents the objectives, background and motivation, data collection methods, quantitative methodology, computational tools, results and analysis, and conclusion related to the study of Liability-Adjusted Bond Portfolio Construction in the Sri Lankan Government Securities Market, based on the Public Debt Report issued by the Central Bank of Sri Lanka (CBSL).

2) Objectives of the Report

The primary objective of this report is to construct a **liability-adjusted bond portfolio using Sri Lankan Treasury securities** which are Treasury bills and bonds. This is particularly relevant for **Sri Lankan insurance companies and pension funds**, which are increasingly required to adopt **Liability-Driven Investment (LDI) strategies** to meet their obligations under tightening **solvency and risk-based capital frameworks** set by the Insurance Regulatory Commission of Sri Lanka (IRCSL). The portfolio construction process adopts a **quantitative approach** utilizing the following techniques:

- 1) **Duration Matching**: This method aligns the **Macauley duration** of the asset portfolio with that of the liabilities, ensuring that the **weighted average time to receipt of cash flows** from government securities matches the timing of expected outflows. In the Sri Lankan context, where long-term instruments such as 10–15-year Treasury bonds are available but illiquid, careful duration matching allows institutions to neutralize the impact of **parallel shifts in interest rates** on the net present value of surplus. This is a foundational technique to minimize interest rate risk while working within the limited maturity spectrum offered in the local market.
- 2) **Convexity Optimization**: While duration matching addresses first order interest rate sensitivity, **convexity** accounts for the curvature in the price-yield relationship. Portfolios with higher convexity experience smaller losses (or larger gains) when interest rates change non linearly. Given that yield curves in Sri Lanka are often irregular and subject to sudden policy-driven shifts, especially during macroeconomic adjustments or IMF-supported fiscal reforms, convexity becomes critical. Optimizing for convexity ensures that even if interest rate changes are large or uneven across maturities, the portfolio remains resilient.

Redington Immunization Criteria: This classical immunization framework strengthens the liability hedge by satisfying three conditions:

- (1) The present value of assets equals the present value of liabilities
- (2) Durations of assets and liabilities are matched
- (3) Asset convexity is greater than or equal to liability convexity.

In the local context, this technique is valuable when insurers model **guaranteed annuity cash flows**, as it ensures surplus stability under small rate shocks. Applying Redington's method helps align investment strategy with **actuarial reserving** requirements in Sri Lanka. (Further explanation in Quantitative methodology section)

Quadratic Programming / Linear Optimization: Given constraints such as **market liquidity, portfolio diversification rules, duration bands, minimum yield targets, and capital allocation limits**, optimization techniques are used to determine the optimal weights of available Treasury securities. **Linear optimization** may be used for basic duration matching, while **quadratic programming** enables trade offs between cost minimization and convexity maximization.

3) Background and Motivation

In Sri Lanka's evolving financial landscape, the growing need for **Liability-Driven Investment (LDI) strategies** among insurance companies and pension funds has become both a regulatory and economic imperative. LDI is a framework in which investment decisions are primarily guided by the **nature, timing, and size of future liabilities**, rather than focusing solely on asset returns. This shift is especially critical for **life insurers and pension schemes** in Sri Lanka, which face long-term, guaranteed payouts such as annuities and retirement benefits. The **primary goal of LDI strategies** is to ensure that asset cash flows are structured to match or exceed liability obligations under different market scenarios, while managing interest rate and reinvestment risks.

The urgency of adopting LDI is reinforced by the regulatory environment set by the **Insurance Regulatory Commission of Sri Lanka (IRCSL)**, which imposes **Risk-Based Capital (RBC) requirements** to ensure that insurers hold sufficient capital to cover adverse liability shocks. Under the RBC framework, insurers must **align their asset portfolios with the nature and risk profile of their liabilities**, particularly under stressed interest rate scenarios. Failure to do so results in increased capital charges or solvency breaches. These solvency pressures are further intensified by Sri Lanka's recent macroeconomic challenges, such as high inflation, currency devaluation, and fluctuating interest rates, which significantly impact the valuation of long-term liabilities and bond assets.

Immunization against **interest rate risk** means the risk that changes in rates will erode the value of assets relative to liabilities. For institutions holding **long-dated liabilities**, even modest rate changes can create large mismatches in present value terms. In the Sri Lankan government securities market, where bond supply is segmented and liquidity is uneven, **constructing an immunized bond portfolio using instruments like Treasury bills and bonds** requires quantitative precision and regulatory awareness. This case study is motivated by the need to provide a **data-driven, regulatory-compliant solution** to Sri Lankan institutions seeking to achieve **interest rate resilience and solvency stability** through structured bond portfolio design.

4) Data Collection

The construction of a liability-adjusted bond portfolio begins with the careful formulation of both the **liability stream** and the **universe of eligible Sri Lankan government securities**. For this case study, the **liability stream** consists of **10 discrete liabilities**, each randomly assigned a value between **LKR 10 million and LKR 100 million**, with **maturities ranging from January 1, 2027, to December 31, 2037**. To realistically reflect institutional discounting behavior, a **lending spread of 50 basis points (0.50% bps)** is added above the yield curve derived from available government bonds, in line with standard asset-liability valuation practice in the Sri Lankan insurance sector.

- **Treasury Bills (T-bills)** with **3-month, 6-month, and 1-year maturities**, which are zero-coupon securities ideal for short-term liability coverage.
- **Treasury Bonds** with tenors of **2, 3, 5, 10, and up to 30 years**, offering semi-annual coupon payments.

For each bond, the following attributes are collected from either real-time or simulated data sourced through the **daily Primary Dealer (PD) reports issued by the Central Bank of Sri Lanka (CBSL)**:

- **Coupon Rate:** Bonds with higher coupon rates provide more periodic cash flows and help enhance portfolio convexity. However, they can also increase reinvestment risk, particularly in a volatile interest rate environment like Sri Lanka's. When selecting bonds, the coupon structure must align with the timing and frequency of liability outflows.

- **Maturity Date:** Maturities should ideally **match or exceed liability due dates** to avoid early redemption risks. In duration and convexity matching, longer maturities are useful for extending portfolio sensitivity, but must be weighed against liquidity constraints in the local market.
- **Prices (Clean / Dirty):**
 - The **clean price** excludes accrued interest and reflects the true market valuation of a bond's principal and coupon profile.
 - The **dirty price** includes accrued interest and is the actual transaction price.
For immunization models and optimization, the **clean price is typically used** for calculating present value and duration, while the dirty price is used when modeling actual cost or executing trades.
- **Yield to Maturity (YTM):** This is the internal rate of return assuming the bond is held to maturity. It is a critical input in calculating **duration and convexity**, and it also serves as a **discount rate** for present value estimation. Bonds with higher YTM's may offer greater return potential but also imply higher risk or lower liquidity.

The selection and calibration of bonds for this portfolio must reflect not just market availability, but also **theoretical alignment with immunization objectives**, such as duration balance, convexity enhancement, and cost efficiency. By leveraging Sri Lanka-specific market data and regulatory considerations, this data-driven approach ensures the portfolio design is both academically robust and practically implementable.

5) Quantitative Methodology

- **Macaulay duration-** measures the **weighted average time** it takes to receive a bond's cash flows, where the weights are based on the present value of each cash flow. It is crucial because if the asset and liability durations are matched, the portfolio becomes **insensitive to small, parallel interest rate shifts**.

$$\text{Duration} = \frac{\sum \frac{t \cdot CF_t}{(1+y)^t}}{\sum \frac{CF_t}{(1+y)^t}}$$

Mathematically, it is calculated as:

where c_t is the cash flow at time t , y is the yield to maturity, and P is the current price of the bond.

- **Convexity-** Captures the **curvature** in the price-yield relationship and measures the bond's sensitivity to **larger or non-linear interest rate changes**. While duration assumes linearity, convexity accounts for

reality but interest rates don't move in straight lines. Its formula is: $\text{Convexity} = \frac{\sum \frac{CF_t \cdot t(t+1)}{(1+y)^{t+2}}}{P}$

Convexity values are always positive for vanilla bonds. It means bond prices increase more than they fall for equal-sized rate movements. A higher convexity implies greater price stability. Negative convexity can occur in bonds with embedded options like callable bonds.

Reasoning Behind Optimization Constraints

The goal of this optimization model is to construct a **cost-efficient bond portfolio** that immunizes future while adhering to sound risk management and regulatory principles. Each constraint serves a specific and necessary purpose:

- **Objective – Minimize total cost:** $\min \sum x_i P_i$

This ensures that the portfolio achieves liability coverage using the **lowest possible market value** of assets (where x_i is the proportion invested in bond i , and p_i is its clean price). For Sri Lankan insurers and pension funds, minimizing cost is critical to **maximize surplus** and meet **capital efficiency** targets under IRCSL's Risk-Based Capital regime.

Duration match:

$$\sum x_i D_i P_i = D_L \cdot PV_L$$

- **Duration Matching Constraint:**

This ensures the **weighted average duration of the asset portfolio** (left side) matches that of the liabilities (right side). It protects the **present value of surplus** from interest rate shifts. This is an essential goal in liability driven investing, especially in Sri Lanka where **interest rate volatility** is high and long duration bonds are limited.

Present value match:

$$\sum x_i P_i = PV_L$$

- **Present Value Matching Constraint:**

This guarantees that the **total value of the bond portfolio equals the present value of liabilities**, ensuring that the assets are fully capable of covering future payments. This constraint reflects **regulatory solvency requirements** and is a fundamental condition for Redington immunization.

Convexity constraint (optional):

$$\sum x_i C_i P_i \geq C_L \cdot PV_L$$

- **Convexity Constraint :**

This adds a second layer of protection by ensuring that the asset portfolio's **convexity** exceeds that of liabilities. It mitigates the effects of **large or non-parallel shifts** in interest rates. In Sri Lanka's context where the yield curve can shift sharply due to macroeconomic events this condition improves portfolio stability and reduces the risk of **surplus erosion**.

Non-negativity:

- **Non-negativity Constraint:** $x_i \geq 0$

This ensures that investment proportions are **realistic and feasible**, disallowing short positions in government securities which is aligned with both **Sri Lankan market norms** and **regulatory investment limits** for insurance and pension institutions.

6) Computational Tools

In this study, Excel Solver is used to find the most suitable combination of bond weights so that the duration of the asset portfolio matches that of the liabilities. To ensure practicality, two key constraints are applied. Those are all bond weights must be greater than or equal to zero and the total weight must add up to 100% which represent a fully invested portfolio.

7) Results and Analysis

Settlement date	5/26/2025	Market shift	0%							
Liabilities	Amount	Maturities	Years to maturity	Yield rate	PV	Weights	Mduration	New value	Time to maturity	
1	\$ 10,091,060.00	2/3/2027	1.70	9.04%	\$ 8,689,613.03	0.1296317	1.6175	\$ 8,689,613.03	618	
2	\$ 10,518,678.00	10/27/2028	3.43	10.31%	\$ 7,458,665.84	0.1112684	3.2531	\$ 7,458,665.84	1250	
3	\$ 11,892,833.00	5/30/2036	11.05	11.91%	\$ 3,326,324.43	0.0496221	10.3921	\$ 3,326,324.43	4022	
4	\$ 12,310,437.00	7/28/2037	12.21	12.04%	\$ 2,966,797.63	0.0442587	11.4829	\$ 2,966,797.63	4446	
5	\$ 12,023,565.00	3/2/2027	1.77	9.09%	\$ 10,273,790.41	0.1532645	1.6921	\$ 10,273,790.41	645	
6	\$ 15,669,237.00	5/22/2035	10.02	11.73%	\$ 5,016,915.73	0.0748424	9.4356	\$ 5,016,915.73	3648	
7	\$ 16,609,183.00	9/6/2036	11.32	11.95%	\$ 4,485,905.34	0.0669208	10.6440	\$ 4,485,905.34	4121	
8	\$ 14,932,483.00	10/8/2033	8.40	11.37%	\$ 5,917,484.57	0.0882771	7.9186	\$ 5,917,484.57	3057	
9	\$ 16,756,607.00	10/7/2036	11.41	11.96%	\$ 4,475,673.60	0.0667681	10.7249	\$ 4,475,673.60	4152	
10	\$ 20,534,684.00	11/27/2028	3.52	10.35%	\$ 14,421,908.35	0.2151461	3.3305	\$ 14,421,908.35	1281	
Sum of PV					\$ 67,033,078.93		5.41	\$ 67,033,078.93	27240	

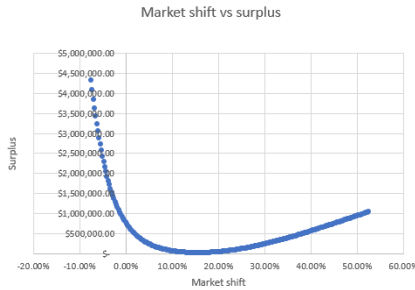
Figure 1:Liabilities of portfolio

Assets										
Bonds and T-Bills	Maturity	Rate	Yield	Price	Mduration	Weights	Value	New price	New value	
Bonds	11.00%2029B	15-Dec-29	11.00%	10.12%	103.13	3.3794	0.28%	\$ 187,501.28	103.15	\$ 187,501.28
	11.00%2030A	15-May-30	11.00%	10.22%	102.99	3.7708	0.00%	\$ -	102.99	\$ -
	11.25%2031A	15-Mar-31	11.25%	10.53%	103.06	4.1328	0.00%	\$ -	103.06	\$ -
	10.00%2031A	15-May-31	18.00%	10.56%	132.34	3.9507	11.91%	\$ 8,073,611.70	132.34	\$ 8,073,611.70
	12.00%2031A	1-Dec-31	12.00%	10.61%	106.43	4.2885	3.57%	\$ 2,422,631.29	106.43	\$ 2,422,631.29
	08.00%2032A	1-Jan-32	8.00%	10.78%	87.09	4.7072	3.87%	\$ 2,625,859.22	87.09	\$ 2,625,859.22
	10.00%2032A	1-Jul-32	18.00%	10.75%	135.31	4.2176	3.52%	\$ 2,388,206.15	135.31	\$ 2,388,206.15
	09.00%2032A	1-Dec-32	9.00%	10.78%	91.08	5.0830	4.14%	\$ 2,808,262.52	91.08	\$ 2,808,262.52
	11.50%2032A	15-Dec-32	11.50%	10.79%	103.60	4.7841	3.89%	\$ 2,663,214.39	103.60	\$ 2,663,214.39
	11.20%2033A	15-Jan-33	11.20%	10.90%	101.50	4.8791	4.00%	\$ 2,709,330.16	101.50	\$ 2,709,330.16
	09.00%2033A	1-Jun-33	9.00%	10.92%	89.94	5.1980	4.22%	\$ 2,864,109.88	89.94	\$ 2,864,109.88
	13.25%2033A	1-Jul-33	13.25%	10.94%	112.17	4.8660	3.99%	\$ 2,702,959.63	112.17	\$ 2,702,959.63
	09.00%2033B	1-Nov-33	9.00%	10.89%	89.72	5.5944	4.51%	\$ 3,056,534.33	89.72	\$ 3,056,534.33
	13.25%2034A	1-Jan-34	13.25%	11.44%	109.72	4.9931	4.08%	\$ 2,764,632.58	109.72	\$ 2,764,632.58
	10.25%2034A	15-Sep-34	10.25%	11.29%	94.05	5.6640	4.56%	\$ 3,090,312.72	94.05	\$ 3,090,312.72
	11.50%2035A	15-Mar-35	11.50%	11.29%	101.20	5.6973	4.58%	\$ 3,106,463.90	101.20	\$ 3,106,463.90
	10.50%2039A	15-Aug-39	10.50%	11.59%	92.44	6.8167	5.38%	\$ 3,649,824.84	92.44	\$ 3,649,824.84
	12.00%2041A	1-Jan-41	12.00%	11.15%	106.22	6.9115	5.45%	\$ 3,695,876.49	106.22	\$ 3,695,876.49
	09.00%2043A	1-Jun-43	9.00%	11.30%	82.46	7.5521	6.00%	\$ 4,066,264.74	82.46	\$ 4,066,264.74
	13.50%2044A	1-Jan-44	13.50%	11.32%	116.76	7.1568	5.63%	\$ 3,814,904.13	116.76	\$ 3,814,904.13
	13.50%2044B	1-Jun-44	13.50%	11.35%	116.60	7.1279	5.61%	\$ 3,800,901.61	116.60	\$ 3,800,901.61
	12.50%2045A	1-Mar-45	12.50%	11.40%	108.50	7.4870	5.86%	\$ 3,975,197.98	108.50	\$ 3,975,197.98
T-Bills	3Months	26-Aug-25	0.00%	7.51%	96.13	0.2449	1.79%	\$ 1,216,223.26	96.13	\$ 1,216,223.26
	6 Months	26-Nov-25	0.00%	7.77%	96.28	0.4814	3.68%	\$ 1,138,618.77	96.28	\$ 1,138,618.77
	1Year	26-May-26	0.00%	8.10%	92.36	0.9611	1.45%	\$ 981,196.84	92.36	\$ 981,196.84
						5.41000	100.00%	\$ 67,033,078.93		\$ 67,033,078.93

Figure 2:Assets of portfolio

The largest weight is allocated to the bond maturing on 15-May-31 (18.00% coupon), contributing 11.91% of the portfolio's value. This selection reflects a preference for high-coupon, medium-term instruments that offer attractive yields (above 10%) while keeping duration aligned with liabilities. Bonds are weighted to collectively match both the present value and the mduration of the liabilities (5.41 years), ensuring optimal immunization under current market conditions. The portfolio's average time to maturity which is 5.41 years. It is aligned with the weighted maturity of the liabilities, indicating that asset cash flows are timed to meet outflows. However, since Sri Lankan government bond liquidity is limited, the actual realization of these flows depends on holding the instruments to maturity, with reinvestment risk mitigated only if rates remain stable.

When we consider sensitivity to interest rate changes, the surplus curve shows a sharp asymmetry. A 100 basis point drop in interest rates generates a significant surplus due to bond price appreciation (~LKR 3–4 million), while a 100 bps increase reduces surplus sharply, nearing breakeven. This convexity is expected, but in the Sri Lankan market where monetary policy shifts are abrupt. This highlights vulnerability to sudden rate hikes. When we consider immunization effectiveness Over Time, as time passes, asset and liability durations will shorten at different rates. Given Sri Lanka's inflation linked rate changes and maturity constraints in the local bond market, periodic monitoring is crucial to prevent mismatches.



Overall, the strategy is sound for current conditions, but its long-term success will depend on active oversight, realistic reinvestment assumptions, and an adaptive approach in response to the volatile macroeconomic environment in Sri Lanka.

8) Conclusion

The liability-matching strategy has achieved precise alignment at the valuation date (26 May 2025), with the present value of assets (LKR 67,033,078.93) exactly equaling the present value of liabilities, and both sides having an identical modified duration of 5.41 years. This reflects strong first-order immunization, which is essential under the IBSL (Insurance Board of Sri Lanka) solvency framework, particularly for long-term insurers who must match the timing and value of future obligations. However, several critical risks and forward-looking considerations remain.

The liability-matching strategy has achieved precise alignment as of 26 May 2025, with the present value of assets (LKR 67,033,078.93) equal to that of liabilities and both having a modified duration of 5.41 years, indicating strong first-order immunization under IBSL solvency norms. However, key risks remain. Reinvestment risk arises as many bonds mature before liabilities, and Sri Lanka's limited long term issuance and rate volatility could force reinvestment at lower yields. Interest rate and illiquidity risks are evident from the surplus vs. market shift graph. A ± 100 bps shift leads to significant surplus volatility, worsened by CBSL driven rate shocks and low secondary market liquidity. Regulatory compliance is currently achieved through proper asset-liability matching, satisfying IBSL guidelines on solvency margins and duration control. But as time progresses, rebalancing will be essential to maintain this alignment, especially given the evolving nature of liabilities and limited local market flexibility.

Future Enhancements

The current model assumes deterministic interest rates, which oversimplifies real-world volatility. Introducing stochastic interest rate models (e.g., CIR or Vasicek) would better capture CBSL-driven rate volatility and improve risk modeling. The liabilities are treated as nominal and fixed. However, many Sri Lankan insurance or pension products are either inflation-indexed or indirectly affected by inflation. Integrating inflation-linked liability modeling would provide a more realistic view of long-term funding adequacy, especially in an economy that has faced double-digit inflation in recent years.

Rebalancing should be explored using scenario analysis, considering Sri Lanka's bond issuance calendar, auction results, and primary dealer access, to build more adaptive reallocation strategies.

Reference

Central Bank of Sri Lanka. (2025, May 26). *[Daily summery report]* [Excel spreadsheet]. Retrieved July 1, 2025, from https://www.cbsl.gov.lk/sites/default/files/cbslweb_documents/about/newreport_20250526.xls

Chen,J.(2022,January7).*Governmentsecurity*.Investopedia.
<https://www.investopedia.com/terms/g/governmentsecurity.asp>